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DLMBIETLT01 EXTRACT, TRANSFORM AND LOAD TECHNOLOGIES

CASE STUDY

HaloGreen – ETL Process for Online Store

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Abstract

This report outlines the design and implementation of an Extract, Transform, Load (ETL) process for HaloGreen, an online store requiring a data warehousing solution to support business intelligence and analytics. The project addresses the challenge of integrating heterogeneous data from three distinct sources: a relational database (sales data), a CSV file (product inventory), and a log file (user web interactions). The selected approach involved choosing a suitable relational database for the core data warehouse (designed in 3rd Normal Form) and associated data marts, and an ETL tool to manage the data pipelines. The ETL process was meticulously designed to extract data from these diverse sources, clean and conform it, apply business rules, perform necessary aggregations (e.g., total revenue per customer/category), and load the data into the central DWH and three specialised data marts (sales, inventory, and customer behaviour). The methodology included prototyping the scenario, documenting all design choices and transformation logic, and creating a comprehensive maintenance guide for the IT department. The result is a robust, scalable data integration framework that transforms raw operational data into structured, analysis-ready information, enabling data-driven decision-making and performance monitoring for the business.

Introduction

HaloGreen is a company operating in the e-commerce sector and currently runs an online store. Being aware of the impact of data in this field, HaloGreen has decided to invest in the development of a small data warehouse. This will not only serve as a central system for assembling the company's data, but it will also be the single source of truth that will transform HaloGreen into a data-driven company where operational and strategic decisions will be made strictly based on data instead of mere intuition. This means that the management will steer the business based on the results of analysis originating from the data stored in this data warehouse. HaloGreen intends to perform several analytics tasks, including aggregations for the calculation of key business metrics like total revenue per customer, total revenue per category, etc. For this reason, the quality and accuracy of the data that will be available in the data warehouse are non-negotiable. However, the management at HaloGreen is aware that an efficient data warehouse strongly depends on the implementation of the extract, transform and load(ETL) process. Thus, it is of utmost importance for HaloGreen that the ET-Process should be robust and consistent. The management has requested its IT department to ensure that the following should be considered during the development of the data warehouse: extensibility to accommodate new data sources, scalability to accommodate company growth, data heterogeneity, performance, and fault tolerance to avoid outage in unexpected events. For the initial development and implementation of the data warehouse, there are three known data sources. These include: a relational database containing sales data, a CSV file containing product inventory and a third source, which is a log file collecting user activities as they interact with the web shop. The data from these sources must be integrated into the data warehouse to allow data analytics operations that will infer insights necessary for making decisions. The rest of this paper is structured as follows. In the technologies section, the chosen tools for the development of the data warehouse will be elaborated, followed by a description of the data. The methodology section will be composed of the data warehouse design, including a brief presentation of its components, which will be followed by the approach detailing the implementation of the extract, load and transformation process. In the approach section, concrete implementations like data ingestion, data normalisation in 3 normal form(3NF) to achieve a consistent data model, and the creation of relational tables in the data warehouse will subsequently be covered. In the fifth section, which is the presentation layer of the data warehouse, data marts will be created to support targeted analysis. The interpretation of results from the analysis will also be presented in this section. In the User manual section, suggestions regarding the maintenance, error handling and improvement of the data warehouse will be provided. In the last section, recommendations and concluding statements will be made stating

what has been learned and how the designed, developed and implemented data warehouse meets the requirements provided by HaloGreen.

Technologies

PostgreSQL

This is an object-oriented relational database management system initiated in 1986 at the University of California in Berkeley, as a component of their POSTGRES project (PostgreSQL Global Development Group, 2025). Lately, this database management system is extensively being developed, and it is free and open source under the PostgreSQL License (PostgreSQL Global Development Group, 2025). Besides being compliant with the ACID principle, it supports Windows, mcOS, Linux, BSD and Solaris operating systems (PostgreSQL Global Development Group, 2025). Programming languages, such as Java, Python, Ruby, Perl, C++, etc., are also supported by PostgreSQL (Simon Lindgren, 2024). The query language in PostgreSQL is SQL with wide support for subqueries, joins, window functions and common table expressions (CTEs) (Yijie Weng & Jianhao Wu, 2025). This DBMS allow the use of custom datatypes as well as primitive data types(e.g Numeric, string, Boolean, Date/Time, etc.), Document (e.g JSON/JSONB), Key-Value(Hstore), etc (PostgreSQL Global Development Group, 2025). As an object-oriented relational database management system, it supports the relational model. This implies that PostgreSQL stores data in relations. As explained in (Lacey-Anne Sanderson et al., 2021), PostgreSQL support different types of indexes, which are data structures that are created on table columns and used to to improve search query performance (Simon Lindgren, 2024). (Microsoft Azure, 2025b) offers a full open-source compatible enterprise level cloud(serverless) Database for PostgreSQL. Currently, the following features are available: Azure AI extensions, Vector search, Autonomous tuning, Vector embeddings, support for distributed PostgreSQL, Extensions, High availability and scaling, Migration services and Microsoft Visual Studio extension.

Azure Data Factory(ADF)

This is a serverless data ingestion/integration service that provides over 90 built-in connectors, freely maintained by (Microsoft Azure, 2025a), to enable data consolidation from different data sources. Azure Data Factory(ADF) is used to easily implement extract, transform and load(ETL) and extract, load and transform (ELT) processes, through its user interface, either by its low-code drag and drop functionality or by writing the desired code (Microsoft Azure, 2025a). To avoid or reduce downtime in rapidly changing environments, it is possible to orchestrate, monitor and manage pipelines in Azure Data Factory.

Power BI

Power BI, among other Microsoft tool stacks, is a robust, versatile business intelligence tool designed to assist businesses to efficiently extract measurable insights from their data(Addepalli Lavanya et al., 2023). It provides multiple data source connectors, enabling users to connect, extract and transform data from various data sources(Addepalli Lavanya et al., 2023). Its user interface facilitates the creation of dashboards and interactive reports through its rich visual capabilities, such as line, pie, doughnut, matrix, map, scatter, and bar charts, providing businesses with a comprehensive presentation of their key performance indicators (Addepalli Lavanya et al., 2023). This tool also provides drill-down and roll-up possibilities to allow users to deep dive into their data. Additionally, Power BI integrates with programming languages like R, Python, and SQL, enabling users to incorporate custom scripts and analytics into their projects(Fehmi Skender & Violeta Manevska, 2022). Besides the Power BI desktop, Power BI service, and Power BI apps are the cloud versions for leveraging self-service at enterprise-level use(Fehmi Skender & Violeta Manevska, 2022). With the ability of Power BI to enhance deep analysis for data-driven decision-making, Power BI is considered a leading choice for business intelligence tasks(Addepalli Lavanya et al., 2023).

Data Description

The data corresponding to the data provided by HaloGreen is generated in Python, specifically Python 3.11.2. There are two Python scripts, namely: interactions.py and iubiprojdata.py(Emmauel Yufenyuy Tardzenyuy, 2025). To ensure reproducibility, seeds were set in the Python scripts. The following are the output files obtained when these scripts are executed and are equally stored in (Emmauel Yufenyuy Tardzenyuy, 2025).

- Customer_interactions.log: This file is produced when interactions.py is executed. Each row is a session, which tracks the amount of time a customer spent on the web shop. It also records all activities a customer made on the web shop within the session. A session can lead to a purchase action, but it must not.
- Customer.csv: This file is produced when iubiprojdata.py is executed. It contains customer master data like first name, last name, date of birth, address, etc. Each row is a unique customer. These are the customers who interact with the web shop.
- Suppliers.csv: This file is created when iubiprojdata.py is executed. It contains supplier information like company name, address, company size, etc. Suppliers are those from whom HaloGreen buys products.
- Inventory.csv: This file is created when iubiprojdata.py is executed. Product information like product name, product category, product subcategory, the current stock level of the product, supplier price and price at which HaloGreen sells each product are recorded

here. Each row is uniquely identified by a product identifier. The price at which HaloGreen sells is at least as high as the price at which HaloGreen buys the product from a supplier.

- Sales_table.csv: This file is generated when iubiprojdata.py is executed. When a customer interacts with the web shop and makes a purchase action, this is recorded in this file. This file contains purchase transaction information like transaction identifier, which identifies every purchase uniquely, product name, customer identifier i.e the customer who has made the purchase, transaction amount calculated as the product of the price and unit(s) of product bought. The total units of products sold cannot exceed the stock level of the inventory.

Methodology

DWH Design

Considering the business requirements, data and aspirations of HaloGreen, the following corresponding relational data warehouse was designed, developed and implemented using the following tools: Three physically separated instances of Azure Databases for PostgreSQL flexible server, Azure Blob Storage, Azure Data Factory and Microsoft Power BI. These tools were reasonably arranged to ensure that their collective interaction will leverage a scalable, performant, fault-tolerant, extendable and loose-coupled system that will fulfil the needs of HaloGreen. In the following, the major components of the data warehouse are briefly presented.

Source Systems

These systems store data generated by operational activities at HaloGreen. Currently, it includes an instance of Azure Database for PostgreSQL flexible server, which is a relational database suitable for storing structured data and an instance of Azure Blob Storage, which leverages containers suitable for storing files.

Staging Area

Data from the source systems is temporarily stored in another instance of Azure Database for PostgreSQL flexible server. The data must be in the form of tables, but not necessarily contain a data model, as is the case in this project. Additionally, this is where the data will be processed or transformed.

Central Data Warehouse

This is composed of another instance of Azure database for PostgreSQL flexible server. The database contains a data model that is developed on the basis of the provided data by HaloGreen. This model guarantees entity and referential integrity to prevent the storage of duplicate entries in the central data warehouse. The storage of duplicate entries could impact the result of data analysis.

Extract-Transform-Load Process

This is a well-thought-out process that involves the extraction of data from the source systems. The extracted data is loaded into the staging area, then transformed or processed before it is loaded into the central data warehouse. This process was developed based on the understanding of the incoming data from the source systems at HaloGreen. For the implementation of this process, Azure Data Factory was used.

The presentation Layer

This is the component of the data warehouse that enables the consumption of the data in the central data warehouse. Different departments, user groups and individuals will import data they require for analysis purposes aimed at supporting management, operational and strategic decision-making. Specifically, Power BI will be used to import data and create data marts to calculate key performance indicators that will be monitored for business steering. Some of the key performance indicators that will be calculated include: total purchase amount(sales or revenue), share of purchase, share of wallet, total revenue per product category, total revenue per product subcategory, total revenue per customer, determine products that are running out of stock, products that have never been sold, monthly website visits, purchase frequency (A. Danailov, 2025; (Virginie BRUNEAU, 2017). Based on these key performance indicators, various dashboards will be developed and deployed in the Power BI service to enable self-service

Fig 1 is a visual representation of the relational data warehouse designed and implemented for HaloGreen.

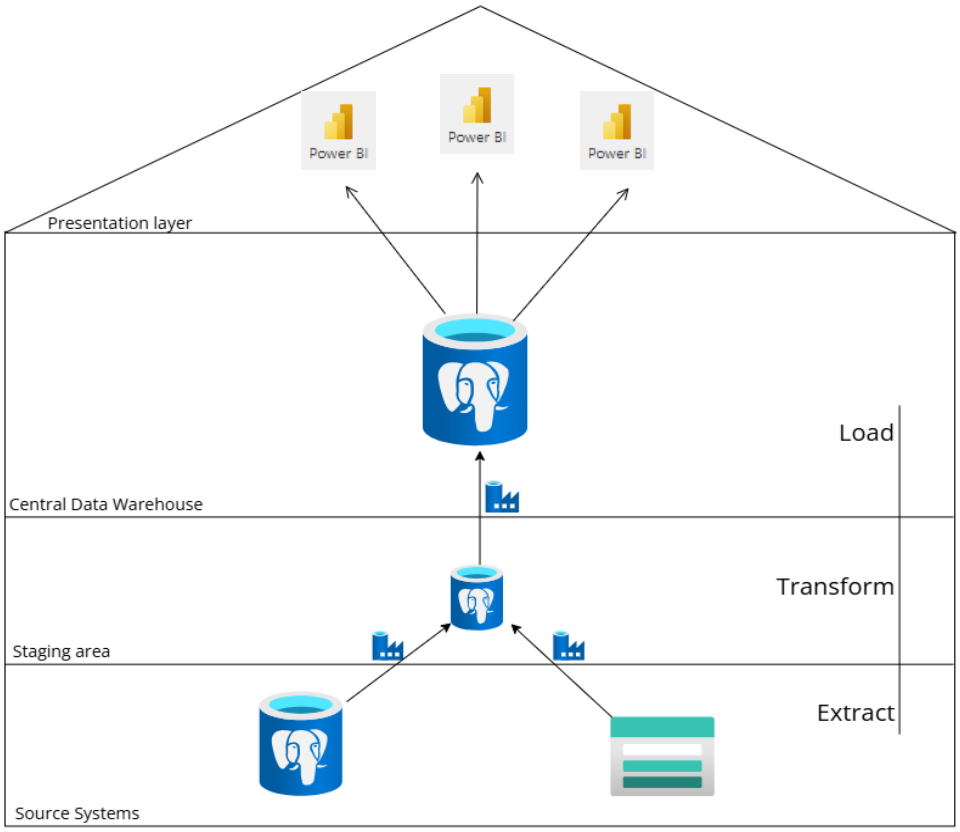


Fig 1:HaloGreen's Relational Data Warehouse (Tardzenyuy Emmanuel Yufenyuy, 2025)

Implementation of a Relational Data Warehouse(RDWH)

Data Upload

At the initial stage, there was no data in any of the source systems. However, based on the description of the tasks, Python was used to generate data that matches the required data described by HaloGreen. The specific files that were generated are: customer_interactions.log, customers.csv, suppliers.csv, inventory.csv and sales_table.csv. Once these files, containing the required data, were generated after running the corresponding Python scripts, the data was manually uploaded into their target source systems. Given that customer_interactions.log contained semi-structured data in JSON format, it was decided that the file be stored in Azure Blob storage, specifically in a container. On the other hand, customers.csv, suppliers.csv, inventory.csv and sales_table.csv were all uploaded in the Azure database for PostgreSQL flexible server transaction system.

Data Staging

After performing data profiling to determine the data that is required for the development of the data warehouse, it was concluded that all the tables and files

currently available in the source systems must be integrated into the data warehouse. However, no data model was developed for the staging area. As a result, corresponding tables of the source systems were created in the staging area without considering relationships that might exist between the tables. The SQL scripts that were used for the creation of the tables in the staging area can be found in (Emmauel Yufenyuy Tardzenyuy, 2025).

Data Modelling

Based on these data sets that must be integrated in the data warehouse, normalisation in three normal form (3NF) was applied to develop the target data model of the central data warehouse. In the development of this data model, primary keys and other constraints like foreign keys and checks were defined. In the implementation of this model with SQL, the primary and foreign keys were used to set entity and referential integrity to avoid the integration of duplicate entries in the central data warehouse. The dwh_tables.slq script in (Emmauel Yufenyuy Tardzenyuy, 2025) was used to implement the data model in the central data warehouse. The following Fig 2 is a visual representation of the data model in the central data warehouse. The primary key fields of each table are in bold. For instance, the primary key of the customer table is customer**_id**.

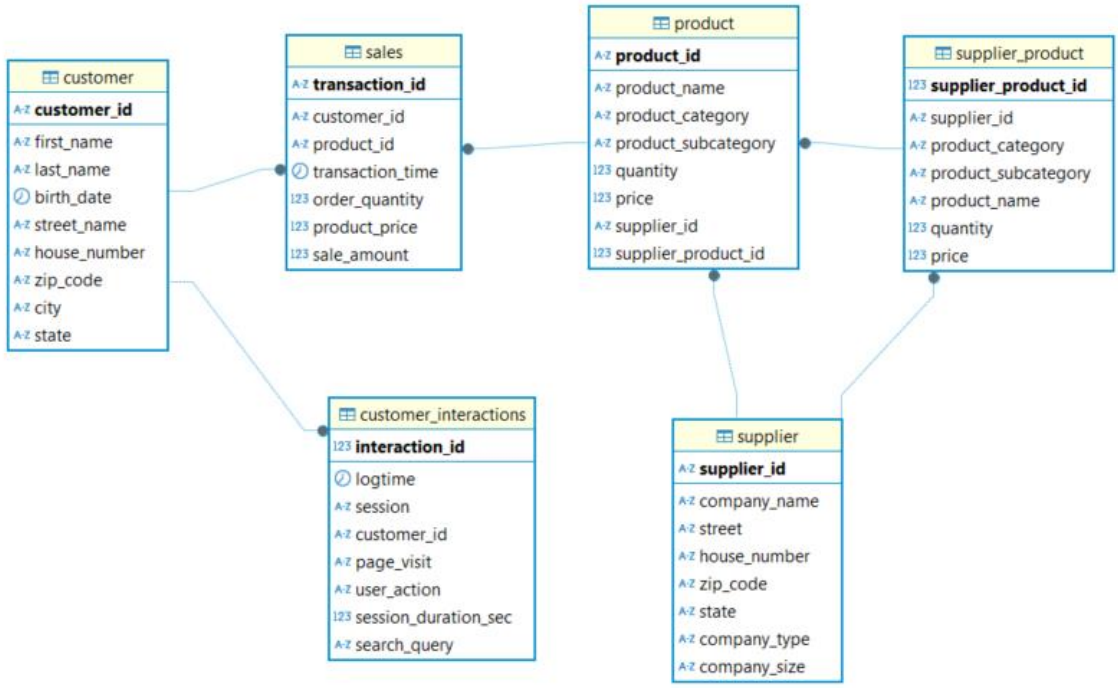


Fig 2:HaloGreen's Central Data Warehouse Data Model in 3 Normal Form (Tardzenyuy Emmanuel Yufenyuy,2025)

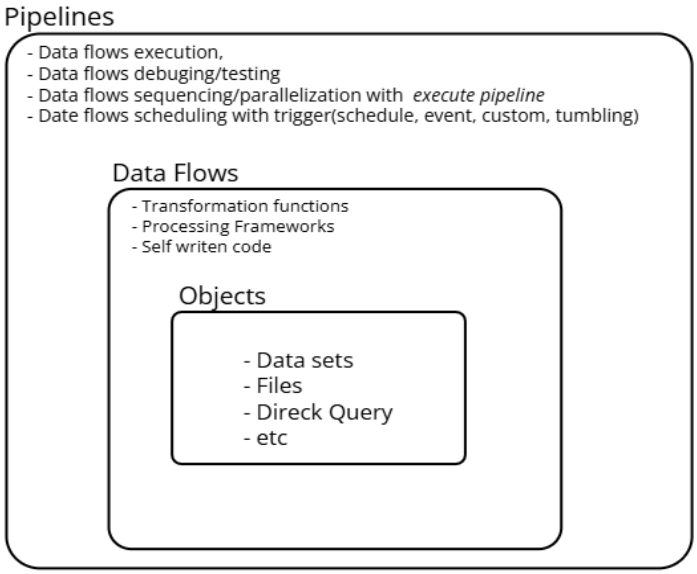


Fig 3:HaloGreen's ETL-Process implementation structure in Azure Data Factory(Tardzenyuy Emmanuel Yufenyuy, 2025).

Azure Data Factory was used in the implementation of the ETL Process. Fig 3 visualises the construct of the ETL process which is described in the following phases.

Extract(E)

In the data extraction phase, a combination of copy data and data flows was used to move data from the two HaloGreen's data sources to their respective data tables in the staging area. During data movement, the keys of the JSON file containing customer interactions with the web were effectively mapped to their corresponding fields on the target table in the staging area. The default data processing setting in Azure Data Factory was not altered.

Transform(T)

Similarly, several data flows were implemented, performing different transformations on the data in the staging area. The specific data transformation and processing steps included:

- Changing the data type of data in the staging to correspond to the data types of the tables in the target data model in the data warehouse.
- Selecting fields from tables in the staging table to correspond to the expected fields of tables in the central data warehouse.
- Performing lookups and altering rows between tables in the staging area and tables in the central data warehouse to determine the delta of data in the staging area that should be moved to the central data warehouse. The delta data can either be data that has changed or new data. The data movement from the staging area to the central data warehouse is via the upsert function provided in Azure Data Factory. With upsert function, the delta can either update existing records in the central data warehouse or insert new records into the central data warehouse, in their respective tables.
- A batch size of 90000 rows is exclusively set for processing customer interaction data with the web since it contains at least 300000 rows. This is set to enhance performance. It was also found that Azure Data Factory offers optimisation options to process data in parallel using partitioning, but this was not explored, given that the current data at HaloGreen could comfortably be processed with the default settings.

Loading(L)

In this final phase of the ETL process, processed and transformed data is loaded into their respective tables in the central data warehouse. During loading, parent or independent tables are first loaded with data before referencing or child tables.

Data Pipelines Orchestration and Testing

For every data flow built, an executable pipeline was linked to it to allow testing in small units. All the pipelines were then indexed and organised in a chronological executable order, considering dependencies. For instance, pipelines A1 and A2 can run in parallel while pipelines C and D can only run in sequence because D depends on C. The organisation of these

pipelines was done in a master pipeline. The successful execution of the master pipeline was tested by scheduling this to run twice. Fig 4 shows a successfully executed master pipeline, while Fig 5 shows the same, including the times of the runs for the individual pipelines, and Fig 6 is a Gantt chart that enables dependencies and parallel pipelines to be visualised.

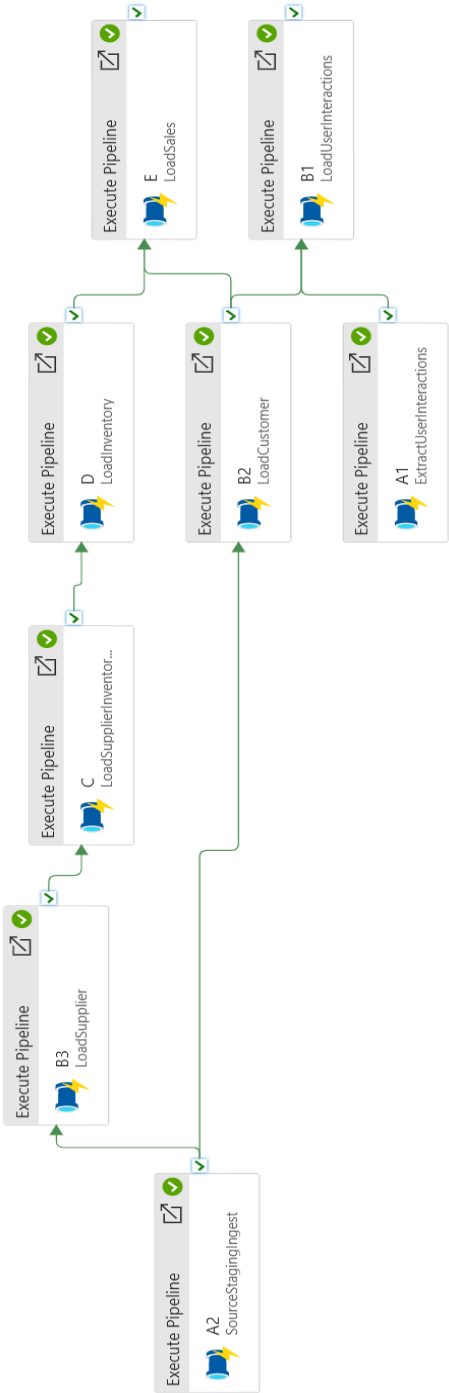
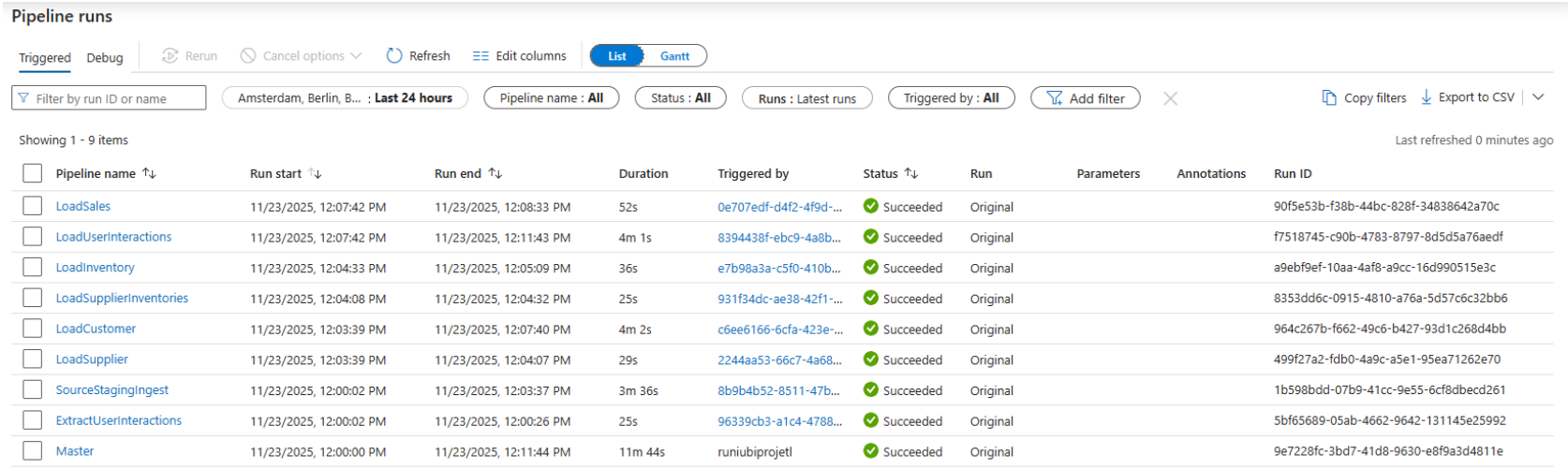


Fig 4:HaloGreen's ETL-Pipeline successful Test(Tardzenyuy Emmanuel Yufenyuy, 2025)



Pipeline name	Run start	Run end	Duration	Triggered by	Status	Run	Parameters	Annotations	Run ID
LoadSales	11/23/2025, 12:07:42 PM	11/23/2025, 12:08:33 PM	52s	0e707edf-d4f2-4f9d-...	Succeeded	Original			90f5e53b-f38b-44bc-828f-34838642a70c
LoadUserInteractions	11/23/2025, 12:07:42 PM	11/23/2025, 12:11:43 PM	4m 1s	8394438f-ebc9-4a8b...	Succeeded	Original			f7518745-c90b-4783-8797-8d5d5a76aedf
LoadInventory	11/23/2025, 12:04:33 PM	11/23/2025, 12:05:09 PM	36s	e7b98a3a-c5f0-410b...	Succeeded	Original			a9ebf9ef-10aa-4af8-a9cc-16d990515e3c
LoadSupplierInventories	11/23/2025, 12:04:08 PM	11/23/2025, 12:04:32 PM	25s	931f34dc-ae38-42f1-...	Succeeded	Original			8353dd6c-0915-4810-a76a-5d57c6c32bb6
LoadCustomer	11/23/2025, 12:03:39 PM	11/23/2025, 12:07:40 PM	4m 2s	c6ee6166-6cfa-423e-...	Succeeded	Original			964c267b-f662-49c6-b427-93d1c268d4bb
LoadSupplier	11/23/2025, 12:03:39 PM	11/23/2025, 12:04:07 PM	29s	2244aa53-66c7-4a68...	Succeeded	Original			499f27a2-fdb0-4a9c-a5e1-95ea71262e70
SourceStagingIngest	11/23/2025, 12:00:02 PM	11/23/2025, 12:03:37 PM	3m 36s	8b9b4b52-8511-47b...	Succeeded	Original			1b598bdd-07b9-41cc-9e55-6cf8dbecd261
ExtractUserInteractions	11/23/2025, 12:00:02 PM	11/23/2025, 12:00:26 PM	25s	96339cb3-a1c4-4788...	Succeeded	Original			5bf65689-05ab-4662-9642-131145e25992
Master	11/23/2025, 12:00:00 PM	11/23/2025, 12:11:44 PM	11m 44s	runiubiprojetl	Succeeded	Original			9e7228fc-3bd7-41d8-9630-e8f9a3d4811e

Fig 5:HaloGreen's ETL Pipeline successful execution in Azure Data Factory(Tardzenyuy Emmanuel Yufenyuy, 2025)

Fig 6:HaloGreen's ETL Pipeline execution Gannt Chart(Tardzenyuy Emmanuel Yufenyuy,2025)

Analysis and Results

Descriptive analysis of sales, inventory and customer behaviour was performed on the data contained in the data marts. This was composed of presenting the calculated measures in the form of visuals and tables. The analysis involved the interpretation of the key performance indicators, and the results include the identification of patterns or correlations, which are vital for decision-making.

Sales

Sales data was imported into Power BI, and a few data processing/transformation steps, like changing data types, eliminating unwanted columns from the imported tables, and creating fact and dimension tables, were made. The required data was connected via their primary and foreign keys to form a data mart used for the analysis of sales transactions. Fig 7 and Fig 8 are visual representations of the data mart and a snapshot of the sales dashboard, respectively.

Overall, HaloGreen has witnessed an increase in sales from 2022 to 2025, with the strongest increase from 2022 to 2023. Indication that the company is doing well. The business experiences a drop in sales from the months of December to February, which could be a result of seasonality. However, there is a sharp increase in sales in the month of March. The total revenue generated amounts to 132 Million Monetary Value, with products in the Audio subcategory accounting for the highest revenue. The full dashboard or report can be accessed via the link provided in (Emmanuel Yufenyuy Tardzenyuy, 2025a).

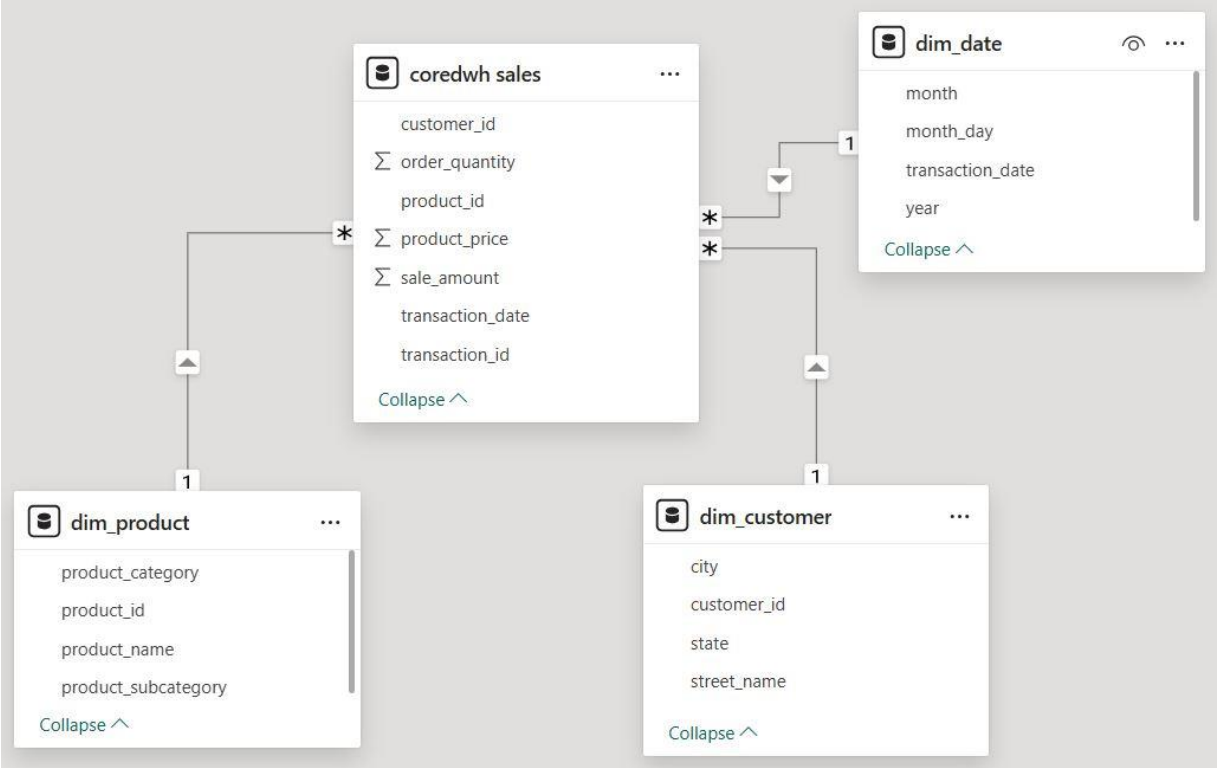


Fig 7:HaloGreen's Sales Data Mart created in Power BI(Tardzenyuy Emmanuel Yufenyuy,2025)

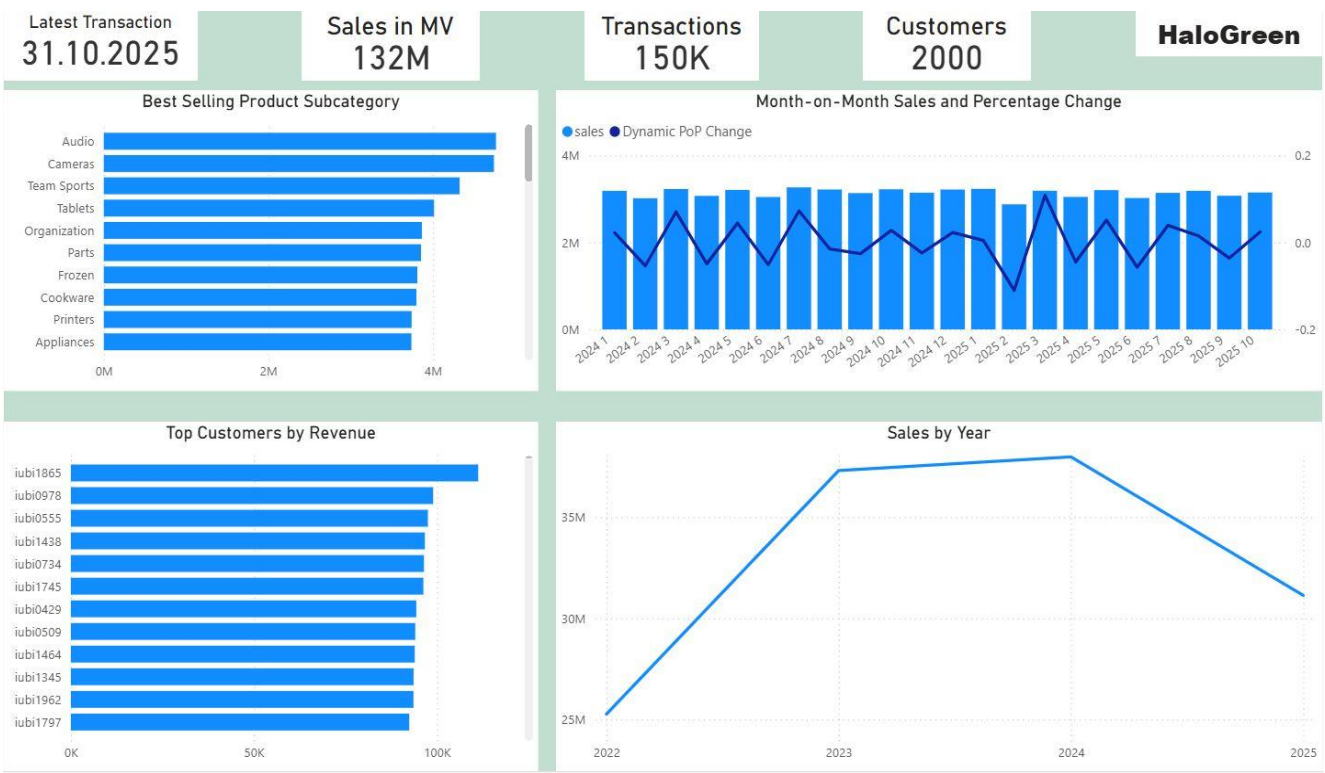


Fig 8:Snapshot of Sales Analysis interactive Dashboard for HaloGreen(Tardzenyuy Emmanuel Yufenyuy,205)

Customer Behaviour

Similarly, a data mart for Customer behaviour data was created. FigX and FigX are visual representations of the data mart and a snapshot of the sales dashboard, respectively.

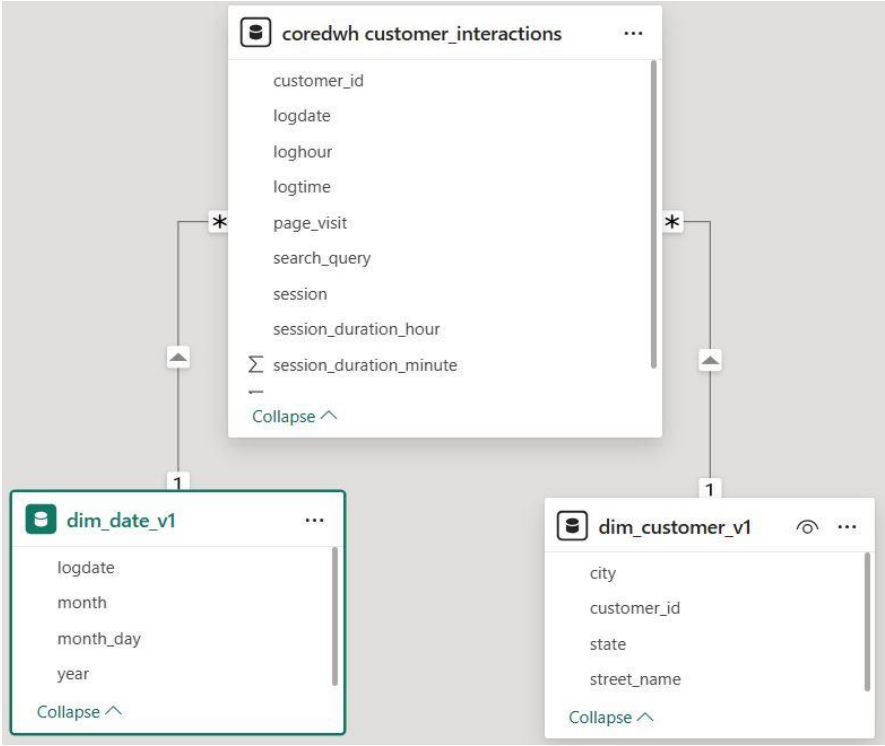


Fig 9:HaloGreen's Customer behaviour Data Mart created in Power BI(Tardzenyuy Emmanuel Yufenyuy, 2025)

Fig 10:Snapshot of Customer behaviour interactive Dashboard for HaloGreen(Tardzenyuy Emmanuel Yufenyuy, 2025)

The number of purchase actions is strongly higher than the number of abandoned carts across several months, with most purchases and cart abandonment recorded in the states of Brandenburg, Niedersachsen and Baden-Württemberg. This indicates a strong and loyal customer base available in these states, suggesting that marketing activities and offers should be directed in these states as compared to other states. There seems to be a positive correlation between hours spent on the web shop and the number of purchase actions; however, this is also observed between hours spent on the online shop and the number of cart abandonments. HaloGreen should consider finding out why customers spend so much time on their website before deciding on the action to take.

Inventory

Similar to the procedure for creating a data mart for customer interactions with the online store, a data mart for inventory was created with a focus on HaloGreen's product data. Fig 9 and Fig 10 are visual representations of the data mart and a snapshot of the sales dashboard, respectively.

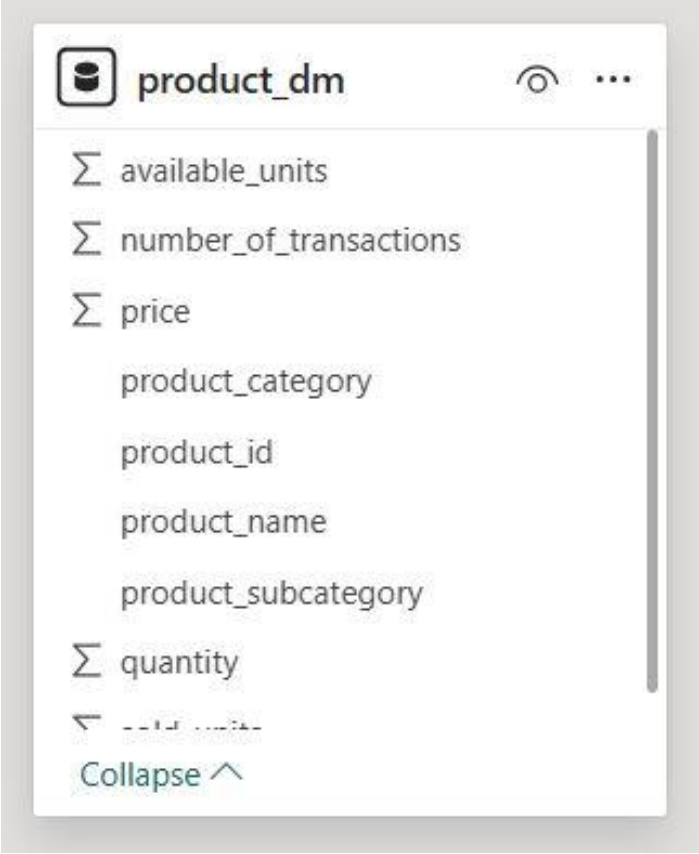
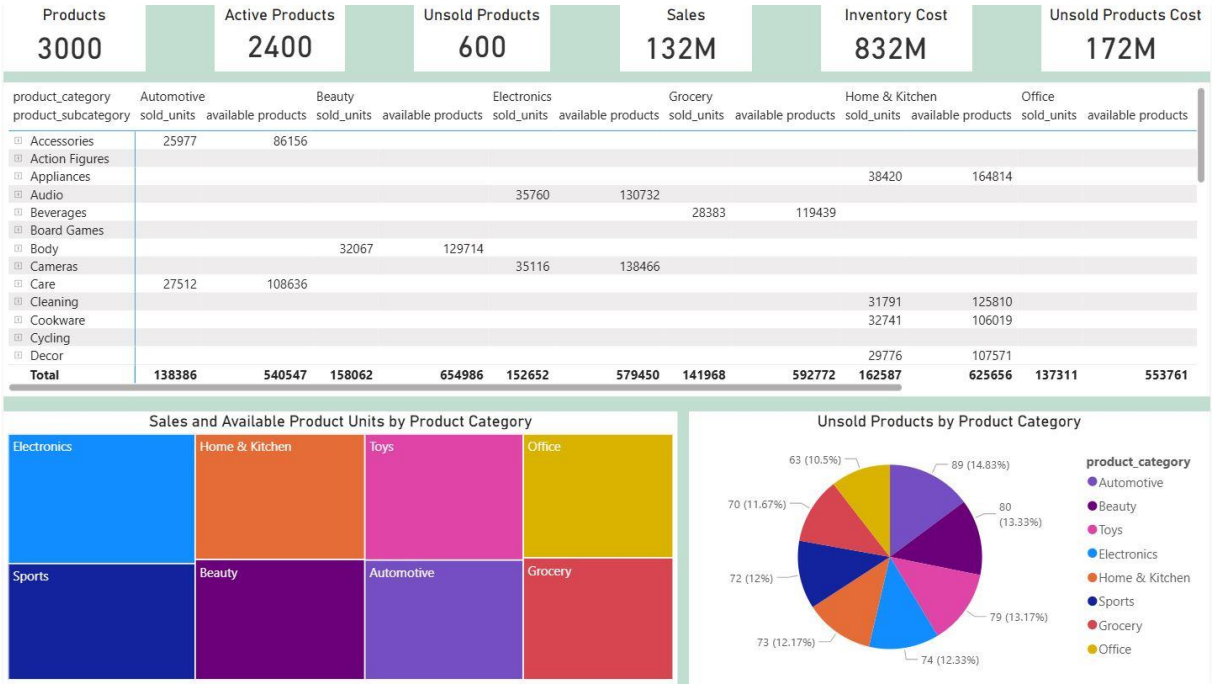


Fig 11: HaloGreen's Inventory Data Mart (Tardzenyuy Emmanuel Yufenyuy,2025)



product_category / product_subcategory	Automotive sold_units	Automotive available products	Beauty sold_units	Beauty available products	Electronics sold_units	Electronics available products	Grocery sold_units	Grocery available products	Home & Kitchen sold_units	Home & Kitchen available products	Office sold_units	Office available products
Accessories	25977	86156										
Action Figures												
Appliances									38420	164814		
Audio					35760	130732						
Beverages							28383	119439				
Board Games												
Body			32067	129714								
Cameras					35116	138466						
Care	27512	108636										
Cleaning									31791	125810		
Cookware									32741	106019		
Cycling												
Decor									29776	107571		
Total	**138386**	**540547**	**158062**	**654986**	**152652**	**579450**	**141968**	**592772**	**162587**	**625656**	**137311**	**553761**

Fig 12:Snapshot of Inventory interactive Dashboard for HaloGreen (Tardzenyuy Emmanuel Yufenyuy, 2025)

According to Fig 10 HaloGreen has a product base of 3000 unique products, with 2400 actively being bought by customers, while 600 products have recorded no sales. The inventory cost for these 3000 products amount to 832 Million Monetary Value of which 132 Million Monetary Value of sales has been recorded corresponding to the sales in Fig 8. The cost of inventory for the 600 products that have recorded no sales amounts to 172 Million Monetary Value. These indicators show that HaloGreen is strongly underperforming, and this is related to the number of available units in stock, for every product category, which is relatively higher than the number of already sold units for every product category. However, the Electronics and Sports product categories have acquired a significant amount of sales as compared to the Office and Grocery product categories. Also, the Automotive and Beauty product categories have the most unsold products as compared to the Office and Grocery product categories

User Manual

As HaloGreen grows with time, it is essential that the data warehouse also evolves to accommodate changes. To achieve this, this data warehouse leverages an ETL process that can integrate several data sources provided by Azure Data Factory (Microsoft Azure, 2025a). It also provides different distributive processing strategies to avoid performance loss when processing large amounts of data. The distributive processing strategy can be specified when creating data flows. Error Logging is equally a great functionality provided by Azure Data Factory when creating data flows. At critical data transformation steps like lookups, log files

can be generated and stored in an Azure Blob container during the execution of the ETL process. This is important for tracking the effective execution of the ETL process and ensuring that the results of transformations are as expected. Log files can also provide information on the amount of time it takes for a transformation to complete. Analysing these logs can form the basis for improving the ETL process. To enable the IT Department at HaloGreen to make these modifications, access rights, for simplicity, exclusively for viewing the implemented ETL, are provided to the tutor of this course. The link to the implemented ETL process in Azure is provided in (Emmanuel Yufenyuy Tardzenyuy, 2025b), and the credentials for login are the tutor's International University email and password. Every change that is made to an existing data flow or newly created data flows in Azure Data Factory should be tested by clicking the debug button. Once the tests are completed, the changes must be published by clicking the publish button in Azure Data Factory to ensure that the latest changes are integrated into the ETL development process. If the changes affect the execution order of the ETL process, then the IT department must make the necessary adjustments to the master pipeline then test this and publish the changes as well. If modifications(e.g creation of new tables) must be made on the data model in the central data warehouse, then the corresponding dwh_tables.sql file used for creating the data model in 3 normal form can be accessed (Emmauel Yufenyuy Tardzenyuy, 2025). Every change made to the model that could affect the ETL process should also be considered, and the necessary adjustments to the ETL process should be made to incorporate these changes.

Recommendations and Conclusion

For effective data warehouse (DWH) management and performance evaluation, the following integrated approach is advised: Ensuring that the user manual is up-to-date with regard to the state of the data warehouse is a foundational step, guaranteeing all internal documentation reflects current system functionality. Simultaneously, providing training to new or existing IT employees at HaloGreen on the structure and functioning of the DWH to ensure they have the necessary skills required when working in a DWH is essential for maintaining operational efficiency and technical proficiency. Regarding performance assessment, HaloGreen should focus externally by comparing HaloGreen's key performance indicators(KPIs) with benchmarks to measure the company's performance against standards in e-commerce, which offers crucial industry context. Internally, success must be measured by evaluating HaloGreen's KPIs, considering business operational and strategic objectives, and ensuring that all data initiatives actively support the company's core mission and strategic goals. Through this, HaloGreen will be able to improve its sales performance, understand its customer base, particularly about the long number of hours spent on its only store before

making final decisions, and finally, control its inventory to focus only on those products that are quickly sold rather than stockpiling with relatively low sales.

In this case study, a comprehensive data management solution was engineered for HaloGreen. A scalable data warehouse was designed and implemented utilising industry-leading tools, specifically configured to meet the company's immediate and future needs. This robust infrastructure, paired with an efficient ETL (Extract, Transform, Load) process, provides HaloGreen with significant operational flexibility. The new system allows for the seamless integration of new data sources, facilitates necessary adjustments to the central data model and ETL pipelines, and supports diverse analytical tasks, all without compromising system performance.

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